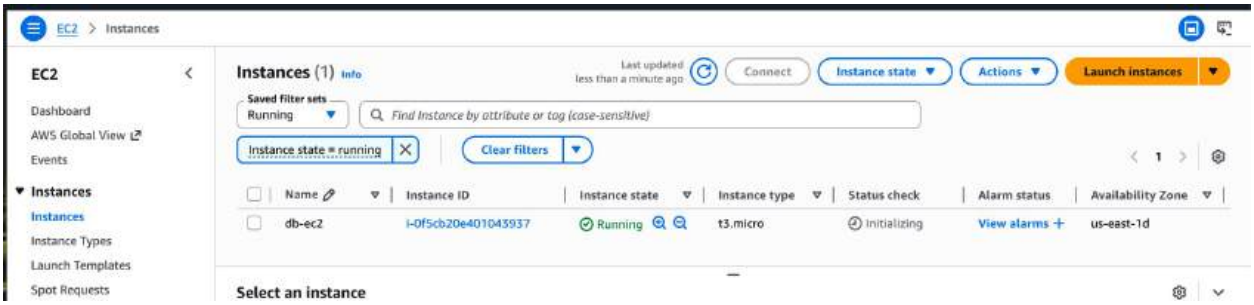
Task on RDS

1. Create MariaDB DB on EC2.

Created an EC2 instance



Using command dnf install mariadb105-server -y

```
Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...
[[root@ip-172-31-0-45 ~]# dnf install mariadb105-server -y
Last metadata expiration check: 0:05:38 ago on Thu Apr 30 09:47:19 2026.
Dependencies resolved.
================================================================================
 Package                    Arch   Version                     Repository   Size
================================================================================
Installing:
 mariadb105-server          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux  10 M
Installing dependencies:
 mariadb-connector-c        x86_64 3.3.10-1.amzn2023.0.1       amazonlinux 211 k
 mariadb-connector-c-config noarch 3.3.10-1.amzn2023.0.1       amazonlinux 9.9 k
 mariadb105                 x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 1.5 M
 mariadb105-common          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux  28 k
 mariadb105-errmsg          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 212 k
 mysql-selinux              noarch 1.0.4-2.amzn2023.0.3        amazonlinux  36 k
 perl-DBD-MariaDB           x86_64 1.22-1.amzn2023.0.4         amazonlinux 153 k
 perl-DBI                   x86_64 1.643-7.amzn2023.0.3        amazonlinux 700 k
 perl-File-Copy             noarch 2.34-477.amzn2023.0.7       amazonlinux  20 k
 perl-Math-BigInt           noarch 1:1.9998.39-2.amzn2023.0.2 amazonlinux 202 k
 perl-Math-BigRat           noarch 0.2624-500.amzn2023.0.2     amazonlinux  42 k
 perl-Math-Complex          noarch 1.59-477.amzn2023.0.7       amazonlinux  46 k
 perl-Sys-Hostname          x86_64 1.23-477.amzn2023.0.7       amazonlinux  16 k
Installing weak dependencies:
 mariadb105-backup          x86_64 3:10.5.29-1.amzn2023.0.1    amazonlinux 6.0 M
```

```
Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...
  Verifying        : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64       18/18

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64
  mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-cracklib-password-check-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
  mysql-selinux-1.0.4-2.amzn2023.0.3.noarch
  perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64
  perl-DBI-1.643-7.amzn2023.0.3.x86_64
  perl-File-Copy-2.34-477.amzn2023.0.7.noarch
  perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch
  perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch
  perl-Math-Complex-1.59-477.amzn2023.0.7.noarch
  perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64

Complete!
[root@ip-172-31-0-45 ~]#
```

Installed mariaDb Server

Using these command we will enable and start the service
systemctl enable mariadb
systemctl start mariadb
yum -y update

```
  perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64

Complete!
[[root@ip-172-31-0-45 ~]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mari
adb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mar
iadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /u
sr/lib/systemd/system/mariadb.service.
[[root@ip-172-31-0-45 ~]# systemctl start mariadb
[[root@ip-172-31-0-45 ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Thu 2026-04-30 09:55:11 UTC; 8s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 27096 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited,>
    Process: 27118 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.ser>
    Process: 27218 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exite>
```

2. Insert some dummy data.

Set Environmental Variables
Setting the admin and password and logging in to MariaDb

DBName=ec2db
DBPassword=admin123456
DBRootPassword=admin123456
DBUser=ec2dbuser
mysqladmin -u root password "${DBRootPassword}"

```
[[root@ip-172-31-0-45 ~]# DBName=ec2db
[[root@ip-172-31-0-45 ~]# DBPassword=admin123456
[[root@ip-172-31-0-45 ~]# DBRootPassword=admin123456
[[root@ip-172-31-0-45 ~]# DBUser=ec2dbuser
[[root@ip-172-31-0-45 ~]# mysql -u root --password="${DBRootPassword}"
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Created a database using CREATE DATABASE ec2db;

```
Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...
ERROR 1049 (42000): Unknown database 'ec2db'
MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
3 rows in set (0.000 sec)

MariaDB [(none)]> CREATE DATABASE ec2db;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| ec2db              |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
4 rows in set (0.000 sec)
```

USE ec2db; → to change to the newly created database

Below commands have been used to create and insert the data into the tables.

CREATE TABLE table1 (id INT, name VARCHAR(45));

INSERT INTO table1 VALUES(1, 'Virat'), (2, 'Sachin'), (3, 'Dhoni'), (4, 'ABD');

SELECT * FROM table1;

```
Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...
MariaDB [(none)]> USE ec2db;
Database changed
MariaDB [ec2db]> CREATE TABLE table1 (id INT, name VARCHAR(45));
Query OK, 0 rows affected (0.009 sec)

MariaDB [ec2db]> NSERT INTO table1 VALUES(1, 'Virat'), (2, 'Sachin'), (3, 'Dhoni
'), (4, 'ABD');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MariaDB server version for the right syntax to use near 'NSE
RT INTO table1 VALUES(1, 'Virat'), (2, 'Sachin'), (3, 'Dhoni'), (4, 'ABD')' at l
ine 1
MariaDB [ec2db]> INSERT INTO table1 VALUES(1, 'Virat'), (2, 'Sachin'), (3, 'Dhon
i'), (4, 'ABD');
Query OK, 4 rows affected (0.002 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [ec2db]> SELECT * FROM table1;
+------+--------+
| id   | name   |
+------+--------+
|    1 | Virat  |
|    2 | Sachin |
|    3 | Dhoni  |
|    4 | ABD    |
```

3. Take the backup of dummy data on EC2.

Created a dump by using mysqldump utility

But before that we need to come out from mariadb to our instance and we need to execute the command on ec2 level so that we can backup on EC2

mysqldump -u root -p ec2db > ec2db_dump.sql

```
[root@ip-172-31-0-45 ~]# mysqldump -u root -p ec2db > ec2db_dump.sql
Enter password:
[root@ip-172-31-0-45 ~]# ls
ec2db_dump.sql
[root@ip-172-31-0-45 ~]#
```

4. Launch MariaDB RDS instance.

FOr that got RDS→Create Database→ Full Configurations →Mariadb→engineversion→Templates→Settings→Name→Credential Settings→Master username→password→Instance config→storage→storage autoscaling(NO) → Availabilty andduarability→Multi AZ Deployment→connectivity(connector **DONT**) → Network type(**IPV4,** Dual) → VPC→DB subnet group → public access(Yes/**NO**) → VPCFirewall → Create

us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#launch-dbinstance:

Aurora and RDS > Databases > Create database

Choose a database creation method

- **Full configuration** — You set all of the configuration options, including ones for availability, security, backups, and maintenance.
- **Easy create** — Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

- Production — Use defaults for high availability and fast, consistent performance.
- Dev/Test — This instance is intended for development use outside of a production environment.
- **Free tier** — Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Settings

Engine version Info

View the engine versions that support the following database features.

▼ Hide filters

Show only versions that support the Amazon RDS Optimized Writes Info

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#launch-dbinstance:

Aurora and RDS > Databases > Create database

Engine version

MariaDB 11.8.6

DB instance identifier Info

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username Info

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

- Managed in AWS Secrets Manager - *most secure* — RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- **Self managed** — Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password Info

Aurora and RDS > Databases > Create database

Master password Info

Password strength Very weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / " @

Confirm master password Info

▼ **Additional credentials settings**

Database authentication options Info

- Password authentication
  Authenticates using database passwords.
- Password and IAM database authentication
  Authenticates using the database password and user credentials through AWS IAM users and roles.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class Info

▼ **Hide filters**

Show instance classes that support Amazon RDS Optimized Writes Info

Aurora and RDS > Databases > Create database

Storage

Storage type Info

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage Info

20 GiB

Allocated storage value must be 20 GiB to 6,144 GiB

▶ **Additional storage configuration**

Availability & durability

Multi-AZ deployment Info

- Create a standby instance (recommended for production usage)
  Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.
- Do not create a standby instance

Connectivity Info

Aurora and RDS > Databases > Create database

Connectivity Info

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- ○ Don't connect to an EC2 compute resource
  Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- ◉ Connect to an EC2 compute resource
  Set up a connection to an EC2 compute resource for this database.

EC2 instance Info

Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0f5cb20e401043937
db-ec2

Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Network type Info

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

- ◉ IPv4
  Your resources can communicate only over the IPv4 addressing protocol.
- ○ Dual-stack mode
  Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) Info

Aurora and RDS > Databases > Create database

Connectivity Info

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- ◉ Don't connect to an EC2 compute resource
  Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- ○ Connect to an EC2 compute resource
  Set up a connection to an EC2 compute resource for this database.

Network type Info

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

- ◉ IPv4
  Your resources can communicate only over the IPv4 addressing protocol.
- ○ Dual-stack mode
  Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-08bc924a2061938df)
3 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Aurora and RDS > Databases > Create database

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access Info

- Yes
  RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- No
  RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- Choose existing
  Choose existing VPC security groups
- Create new
  Create new VPC security group

Existing VPC security groups

Choose one or more options

default

Availability Zone Info

us-east-1a

Aurora and RDS > Databases > Create database

Storage

Storage type Info

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage Info

20 GiB

Allocated storage value must be 20 GiB to 6,144 GiB

▼ Additional storage configuration

Storage autoscaling Info

Provides dynamic scaling support for your database's storage based on your application's needs.

- Enable storage autoscaling
  Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Availability & durability

Multi-AZ deployment Info

- Create a standby instance (recommended for production usage)
  Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.

Aurora and RDS > Databases > Create database

▼ **Additional configuration**

Database port Info

TCP/IP port that the database will use for application connections.

3306

Tags - *optional*

A tag consists of a case-sensitive key-value pair.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Monitoring Info

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights** pricing is separate from RDS monthly estimates. See Amazon CloudWatch pricing.

Database Insights - Advanced

- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

Database Insights - Standard

Aurora and RDS > Databases > Create database

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

☐ Audit log

☐ Error log

☐ General log

☐ iam-db-auth-error log

☐ Slow query log

IAM role

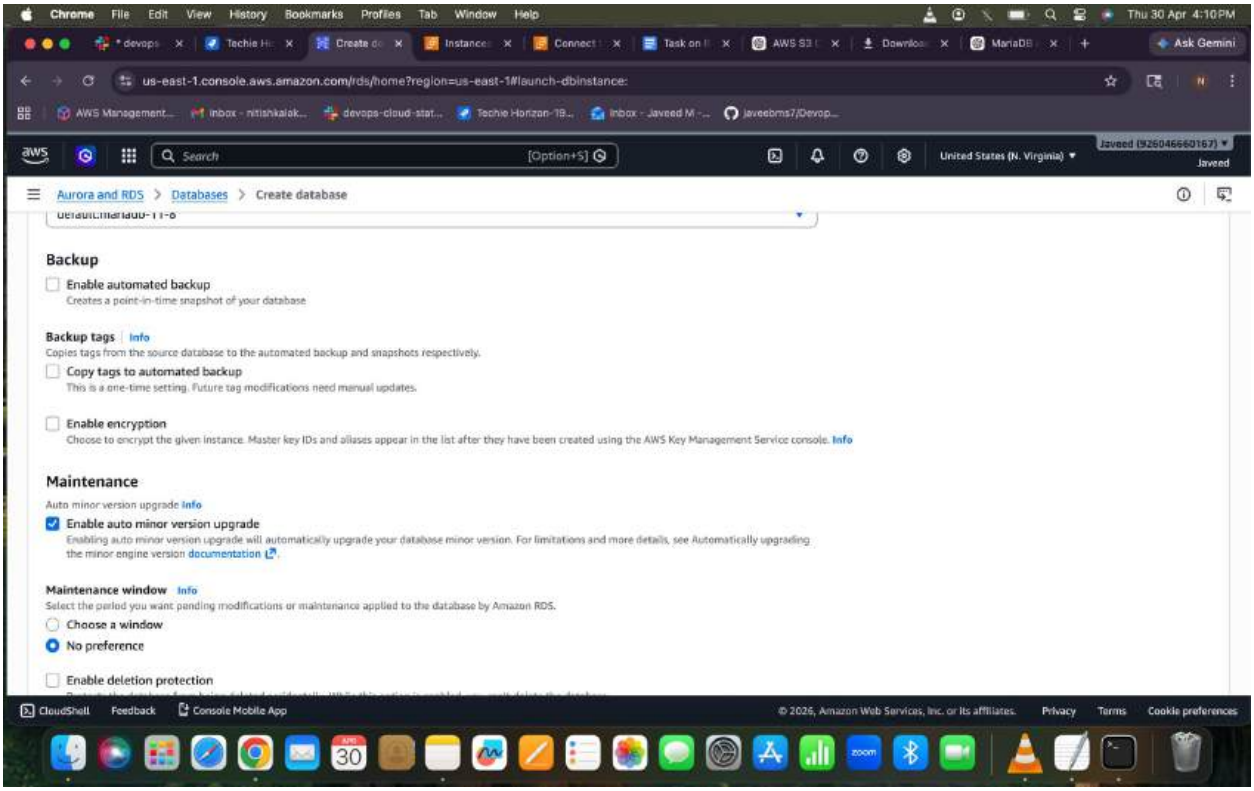
The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

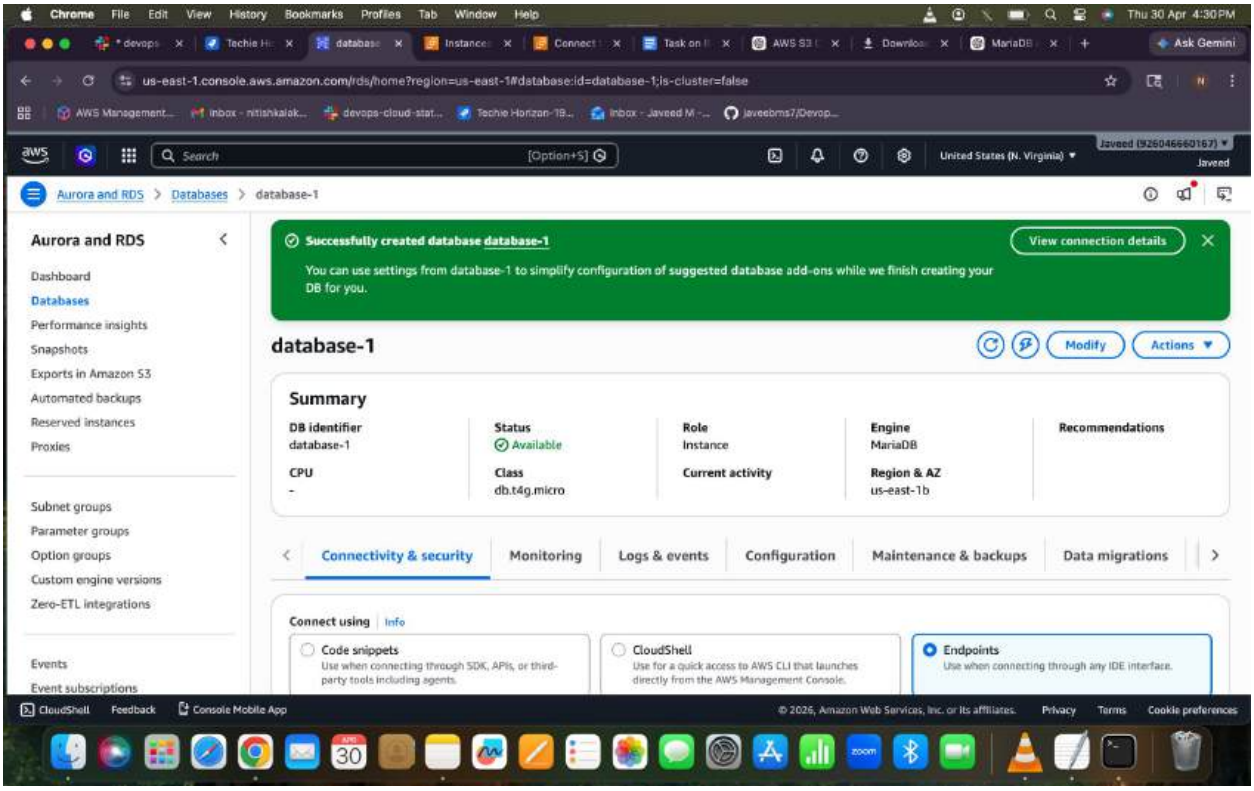
▶ **Additional configuration**

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

ⓘ You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.



Database instance has been successfully created.



We need to add port in the security group of our instance

Data Base is available now use the command
mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl
Ssl isused forsecuring theconnection

```
[root@ip-172-31-0-45 ec2-user]# mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.a
mazonaws.com -P 3306 -u admin -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.8.6-MariaDB managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

We can see what what databases are their by using SHOW DATABASES; from our ec2 instances

```
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 128
Server version: 11.8.6-MariaDB managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES
    -> \c
MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.006 sec)

MariaDB [(none)]>
```

5. Migrate database from EC2 to RDS.
   Will create a backup database and use the mysql command with where the data needs to go to get the database from ec2→rds

```
Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@ec2-3-90-224-229.compute-1.amazonaws.com — 204×54

MariaDB [(none)]> SHOW DATABASE;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'DATABASE' at line 1
MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| ec2db              |
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.001 sec)

MariaDB [(none)]> mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl ec2db > ec2db_dump.sql
    -> \c
MariaDB [(none)]>
[5]+  Stopped                 mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl
[ec2-user@ip-172-31-0-45 ~]$ sudo su
[root@ip-172-31-0-45 ec2-user]# cd
[root@ip-172-31-0-45 ~]# mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl ec2db <  ec2db_dump.sql
Enter password:
[root@ip-172-31-0-45 ~]# ls
ec2db_dump.sql
[root@ip-172-31-0-45 ~]# mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 173
Server version: 11.8.6-MariaDB managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE ec2db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [ec2db]> SELECT * FROM tabel1;
ERROR 1146 (42S02): Table 'ec2db.tabel1' doesn't exist
MariaDB [ec2db]> SELECT * FROM table1;
+------+--------+
| id   | name   |
+------+--------+
|    1 | Virat  |
|    2 | Sachin |
|    3 | Dhoni  |
|    4 | ABD    |
+------+--------+
4 rows in set (0.002 sec)
```

6. Install MySQL DB on EC2.

Stop existing MariaDB service
Uninstall MariaDB packages from EC2 instance
Remove existing database directories and configuration files
Clear DNF/YUM cache to avoid conflicts
Add MySQL official repository to the system
Install MySQL Community Server package
Start MySQL service (mysqld)
Verify MySQL service status is active and running
Retrieve temporary root password from logs
Run secure installation (mysql_secure_installation)
Set root password and apply security configurations
Validate installation by connecting to MySQL
Confirm MySQL version and service functionalit

```
[[root@ip-172-31-0-45 ~]# dnf list installed | grep mysql
[[root@ip-172-31-0-45 ~]# sudo dnf install -y https://dev.mysql.com/get/mysql80-c
ommunity-release-el9-1.noarch.rpm
Amazon Linux 2023 repository                      61 MB/s |  59 MB     00:00
Amazon Linux 2023 Kernel Livepatch repository    274 kB/s |  31 kB     00:00
mysql80-community-release-el9-1.noarch.rpm        45 kB/s |  10 kB     00:00
Dependencies resolved.
================================================================================
 Package                        Arch        Version     Repository         Size
================================================================================
Installing:
 mysql80-community-release      noarch      el9-1       @commandline       10 k

Transaction Summary
================================================================================
Install  1 Package
```

Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...

```
Transaction Summary
================================================================================
Install  1 Package

Total size: 10 k
Installed size: 5.7 k
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : mysql80-community-release-el9-1.noarch                 1/1
  Verifying        : mysql80-community-release-el9-1.noarch                 1/1

Installed:
  mysql80-community-release-el9-1.noarch

Complete!
[root@ip-172-31-0-45 ~]# sudo dnf install -y mysql-community-server --nogpgcheck
MySQL 8.0 Community Server                        58 MB/s | 3.3 MB     00:00
MySQL Connectors Community                       7.6 MB/s | 114 kB     00:00
MySQL Tools Community                             47 MB/s | 1.6 MB     00:00
```

Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...

```
Complete!
[root@ip-172-31-0-45 ~]# sudo dnf install -y mysql-community-server --nogpgcheck
MySQL 8.0 Community Server                      58 MB/s | 3.3 MB     00:00
MySQL Connectors Community                     7.6 MB/s | 114 kB     00:00
MySQL Tools Community                           47 MB/s | 1.6 MB     00:00
Dependencies resolved.
================================================================================
 Package                          Arch    Version       Repository          Size
================================================================================
Installing:
 mysql-community-server           x86_64  8.0.46-1.el9  mysql80-community   50 M
Installing dependencies:
 mysql-community-client           x86_64  8.0.46-1.el9  mysql80-community  3.5 M
 mysql-community-client-plugins   x86_64  8.0.46-1.el9  mysql80-community  1.4 M
 mysql-community-common           x86_64  8.0.46-1.el9  mysql80-community  558 k
 mysql-community-icu-data-files   x86_64  8.0.46-1.el9  mysql80-community  2.3 M
 mysql-community-libs             x86_64  8.0.46-1.el9  mysql80-community  1.5 M

Transaction Summary
================================================================================
Install  6 Packages

Total download size: 59 M
Installed size: 337 M
```

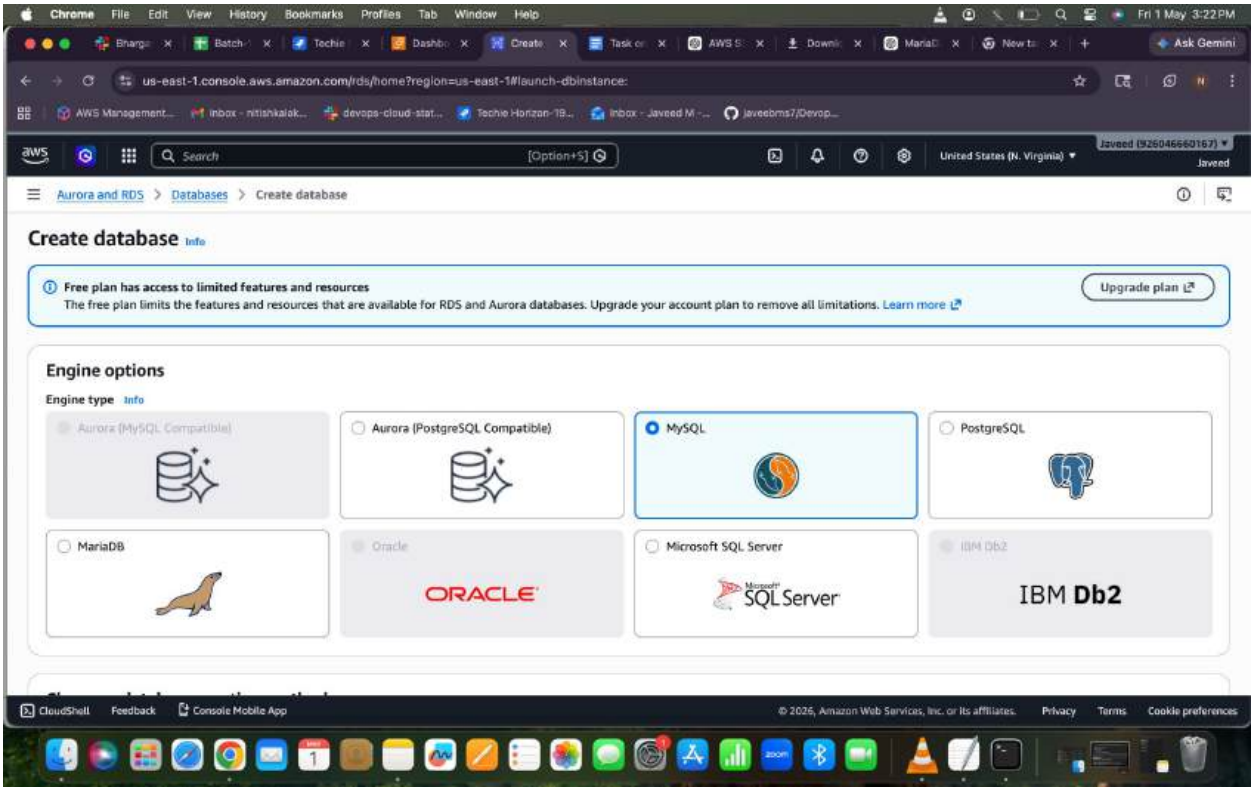
Downloads — root@ip-172-31-0-45:~ — ssh -i newpem.pem ec2-user@e...

```
Installed:
  mysql-community-client-8.0.46-1.el9.x86_64
  mysql-community-client-plugins-8.0.46-1.el9.x86_64
  mysql-community-common-8.0.46-1.el9.x86_64
  mysql-community-icu-data-files-8.0.46-1.el9.x86_64
  mysql-community-libs-8.0.46-1.el9.x86_64
  mysql-community-server-8.0.46-1.el9.x86_64

Complete!
[[root@ip-172-31-0-45 ~]# sudo systemctl start mysqld
[[root@ip-172-31-0-45 ~]# systemctl status mysqld
● mysqld.service - MySQL Server
     Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: d>
     Active: active (running) since Thu 2026-04-30 13:23:41 UTC; 10s ago
       Docs: man:mysqld(8)
             http://dev.mysql.com/doc/refman/en/using-systemd.html
    Process: 34459 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, statu>
   Main PID: 34574 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 1014)
     Memory: 416.1M
        CPU: 4.449s
     CGroup: /system.slice/mysqld.service
             └─34574 /usr/sbin/mysqld
```

7. Launch MySQL RDS image

FOr that got RDS→Create Database→ Full Configurations →MYSQL→engineversion→Templates→Settings→Name→Credential Settings→Master username→password→Instance config→storage→storage autoscaling(NO) → Availabilty andduarability→Multi AZ Deployment→connectivity(connector **DONT**) → Network type(**IPV4,** Dual) → VPC→DB subnet group → public access(Yes/**NO**) → VPCFirewall → Create



Chrome File Edit View History Bookmarks Profiles Tab Window Help Fri 1 May 3:23 PM

us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#launch-dbinstance:

Aurora and RDS > Databases > Create database

Availability and durability

Deployment options Info

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the Amazon RDS service level agreement (SLA).

Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency

Write/read endpoint — AZ 1 — Primary instance + SSD
Reader endpoints — AZ 2 — Readable standby + SSD; AZ 3 — Readable standby + SSD

Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
- 99.95% uptime
- Redundancy across Availability Zones

Write/read endpoint — AZ 1 — Primary instance
Standby (no endpoint) — AZ 2 — Standby

Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:
- 99.5% uptime
- No data redundancy

Write/read endpoint — AZ 1 — Primary instance

Settings

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us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#launch-dbinstance:

Aurora and RDS > Databases > Create database

Engine version

MySQL 8.4.8

☐ **Enable RDS Extended Support** Info
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the RDS for MySQL documentation.

DB instance identifier Info
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-2

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username Info
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

- ○ Managed in AWS Secrets Manager - *most secure*
  RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- ◉ Self managed
  Create your own password or have RDS create a password that you manage.

☐ Auto generate password

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Aurora and RDS > Databases > Create database

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class Info

▼ Hide filters

- Show instance classes that support Amazon RDS Optimized Writes Info
  Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
- Include previous generation classes

- Standard classes (includes m classes)
- Memory optimized classes (includes r and x classes)
- Burstable classes (includes t classes)

Instance type

db.t4g.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Aurora and RDS > Databases > Create database

Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage Info

20 GiB

Allocated storage value must be 20 GiB to 6,144 GiB

▶ Additional storage configuration

Connectivity Info

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- Don't connect to an EC2 compute resource
  Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- Connect to an EC2 compute resource
  Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-08bc924a2061938df)
3 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

Aurora and RDS > Databases > Create database

After a database is created, you can't change its VPC.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-08bc924a2061938df
5 Subnets, 2 Availability Zones

Public access Info

- Yes
  RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- No
  RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- Choose existing
  Choose existing VPC security groups
- Create new
  Create new VPC security group

Existing VPC security groups

Choose one or more options

Aurora and RDS > Databases > Create database

Existing VPC security groups

Choose one or more options

default

Availability Zone Info

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

- Create an RDS Proxy Info
  RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.

Certificate authority - *optional* Info

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

▸ **Additional configuration**

Tags - *optional*

Aurora and RDS > Databases > Create database

Monitoring Info

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights** pricing is separate from RDS monthly estimates. See Amazon CloudWatch pricing.

- Database Insights - Advanced
  - Retains 15 months of performance history
  - Fleet-level monitoring
  - Integration with CloudWatch Application Signals
- Database Insights - Standard

Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

- [ ] Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- [ ] Audit log
- [ ] Error log
- [ ] General log
- [ ] iam-db-auth-error log
- [ ] Slow query log

Aurora and RDS > Databases > Create database

Option group Info

default:mysql-8-4

Backup

- [ ] Enable automated backup

Creates a point-in-time snapshot of your database

Backup tags | Info

Copies tags from the source database to the automated backup and snapshots respectively.

- [ ] Copy tags to automated backup

This is a one-time setting. Future tag modifications need manual updates.

- [ ] Enable encryption

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. Info

Maintenance

Auto minor version upgrade Info

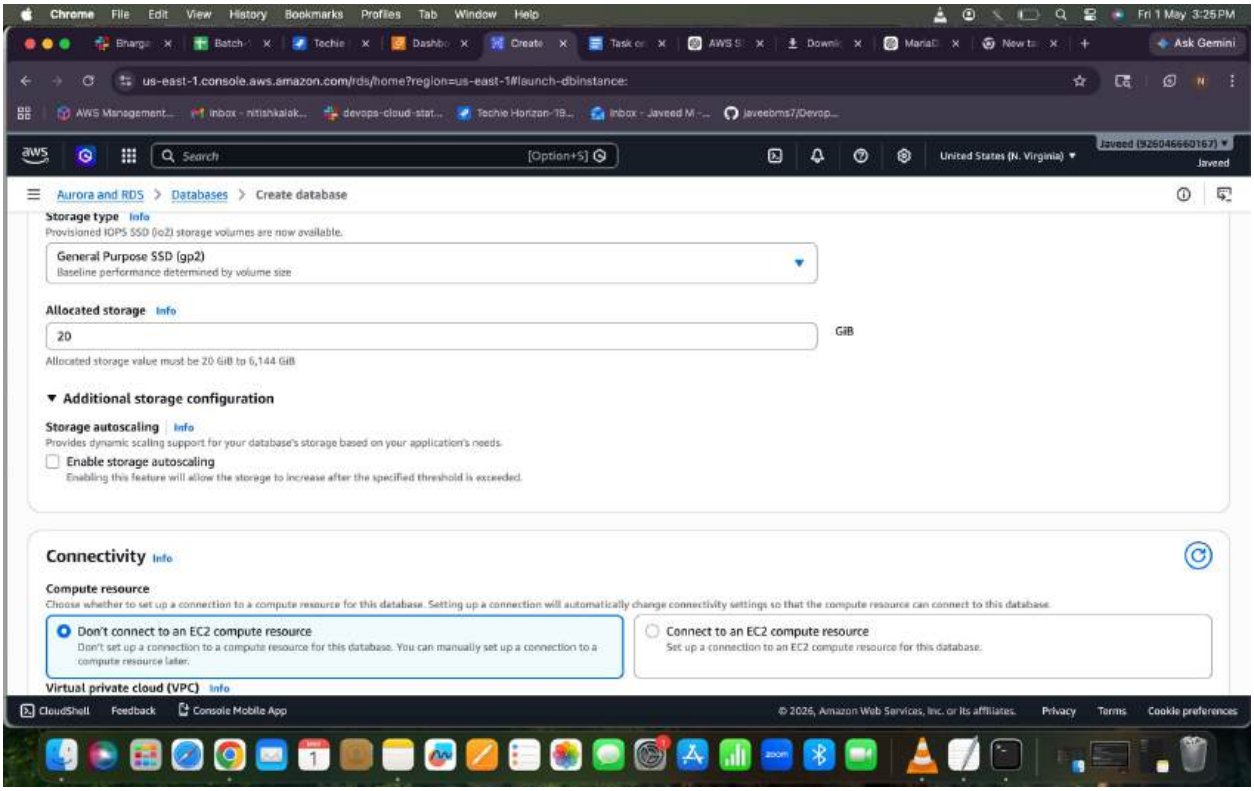
- [x] Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version documentation.

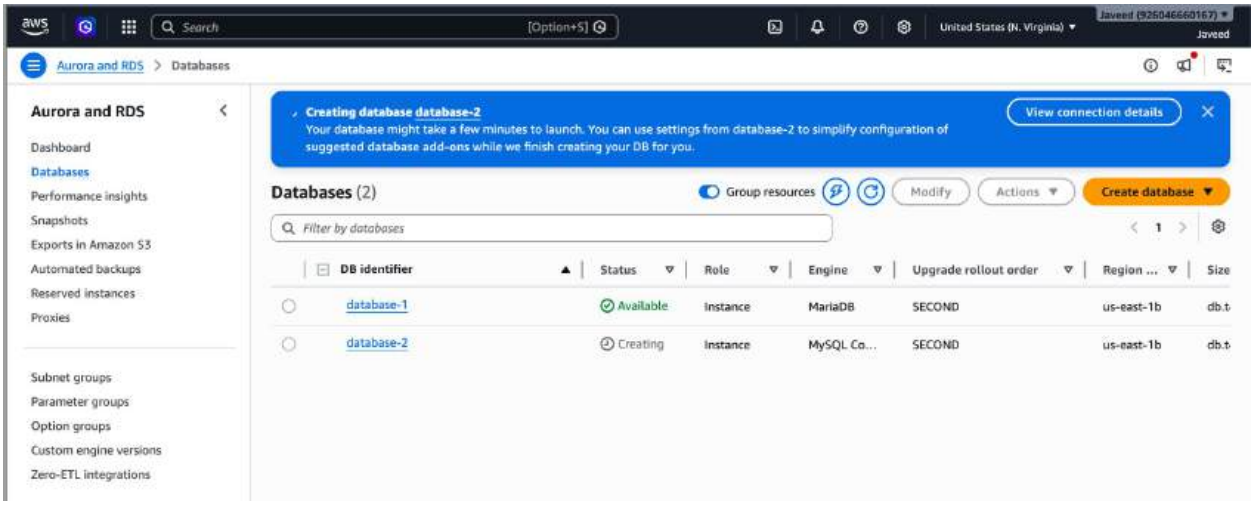
Maintenance window Info

Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

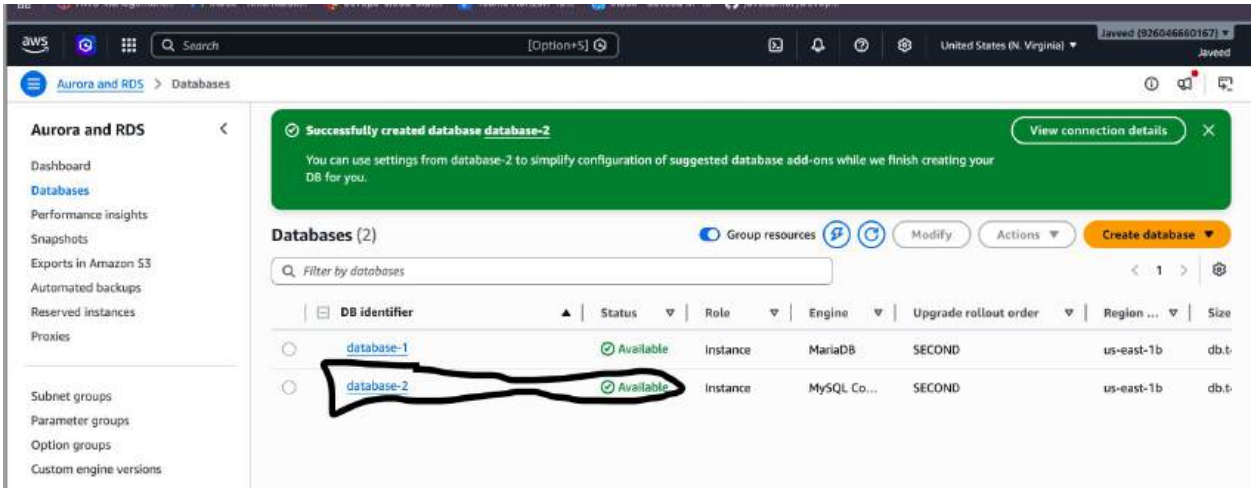
- ( ) Choose a window



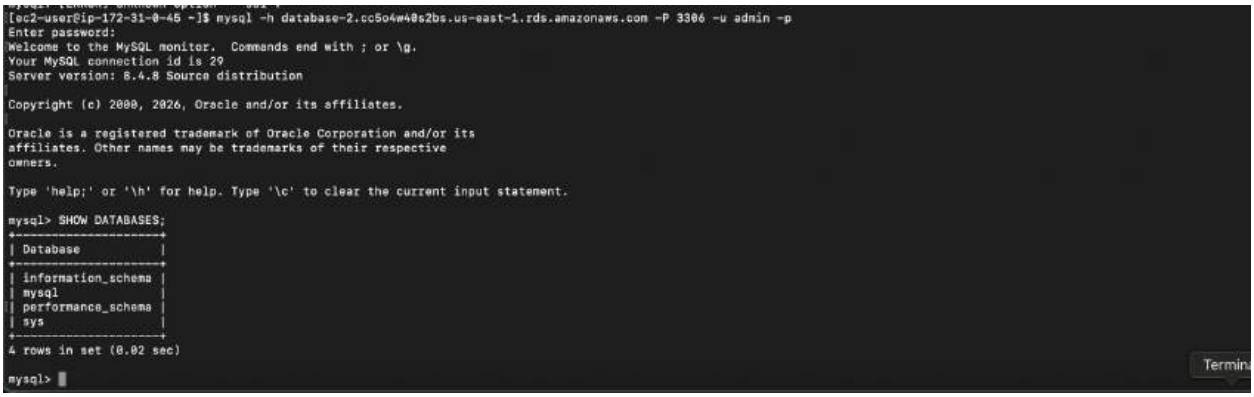
Creating a Mysql Database



Mysql Database created



We need to add port in the security group of our instance



Data Base is available now use the command
mysql -h database-1.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p to connect to sql server from the instance.
After connecting to the sql server just check if everything is working fine by checking the database We can see what what databases are their by using SHOW DATABASES; from our ec2 instances

```
[ec2-user@ip-172-31-0-45 ~]$ mysql -h database-2.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 29
Server version: 8.4.8 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.02 sec)

mysql>
```
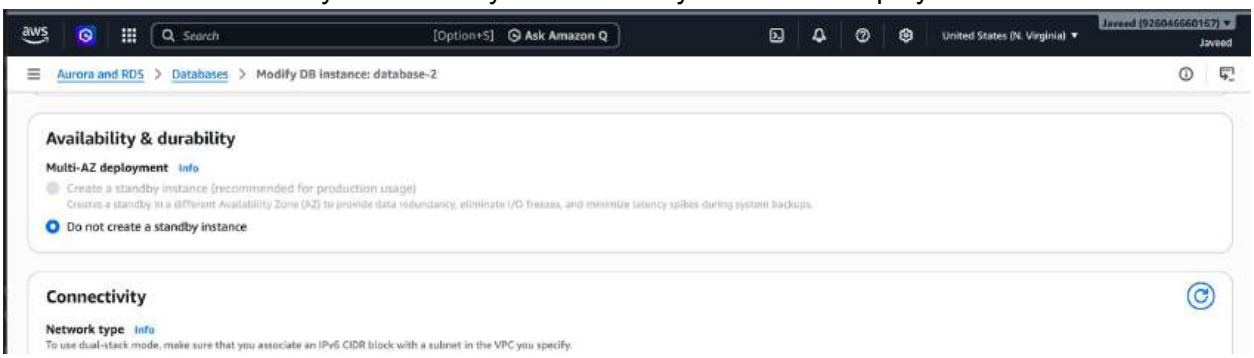

8. Configure Multi-AZ.

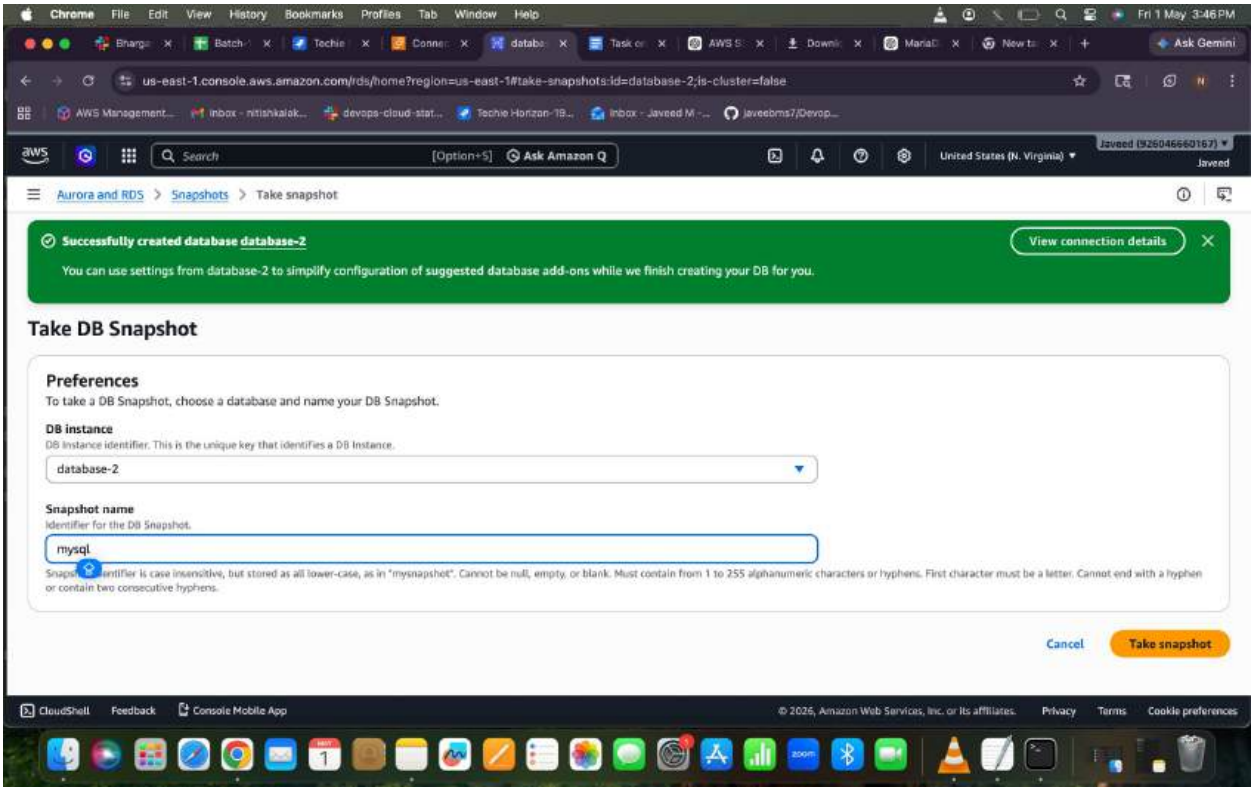
Tried modifying the existing database to multi AZ bus as i have been using free-tier it isnt allowing me to do.

RDS→Database→Modify→Availability and Durability→Multi AZ DEployment→Create

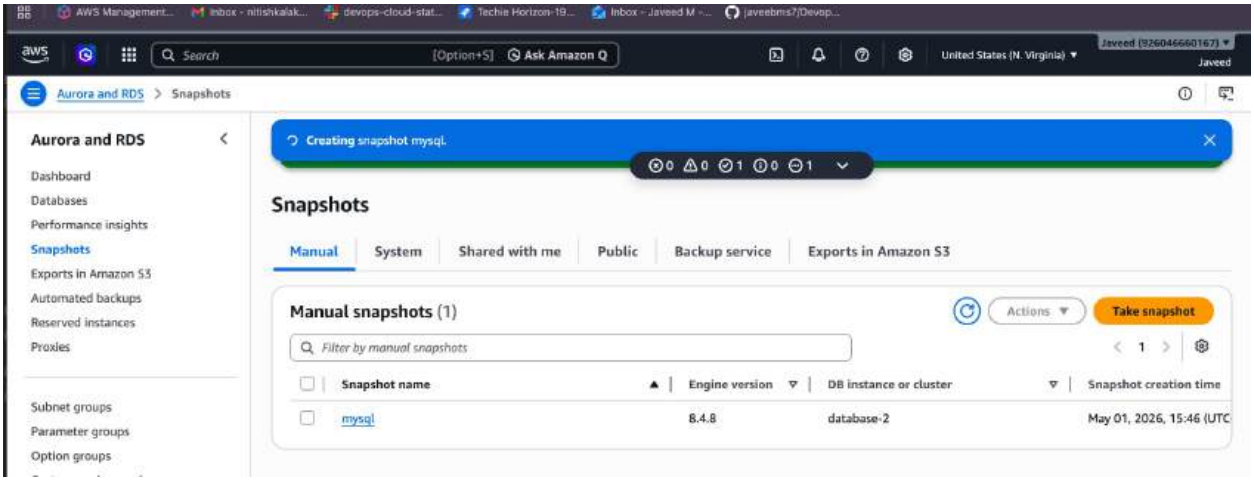


9. Take backup of DB and restore the DB.

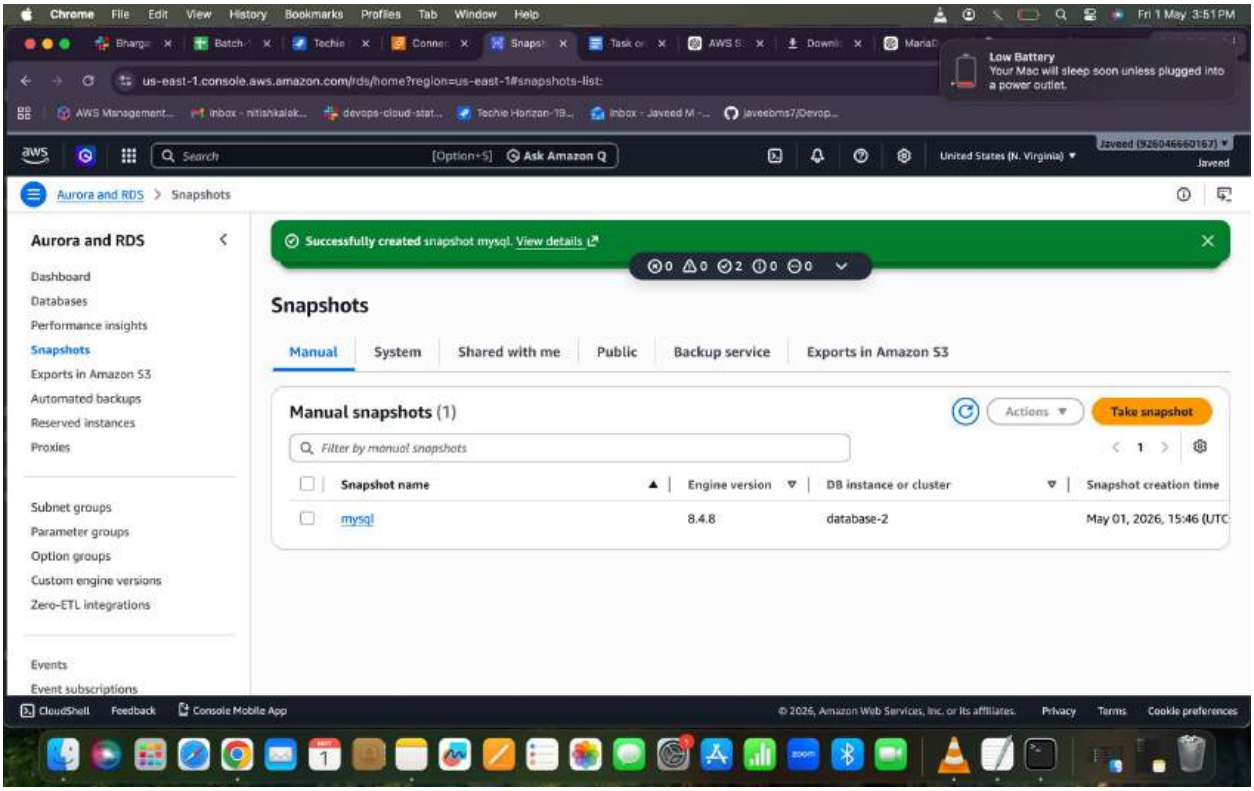
RDS→Database→Actions→Snapstop→name→ create snapshot



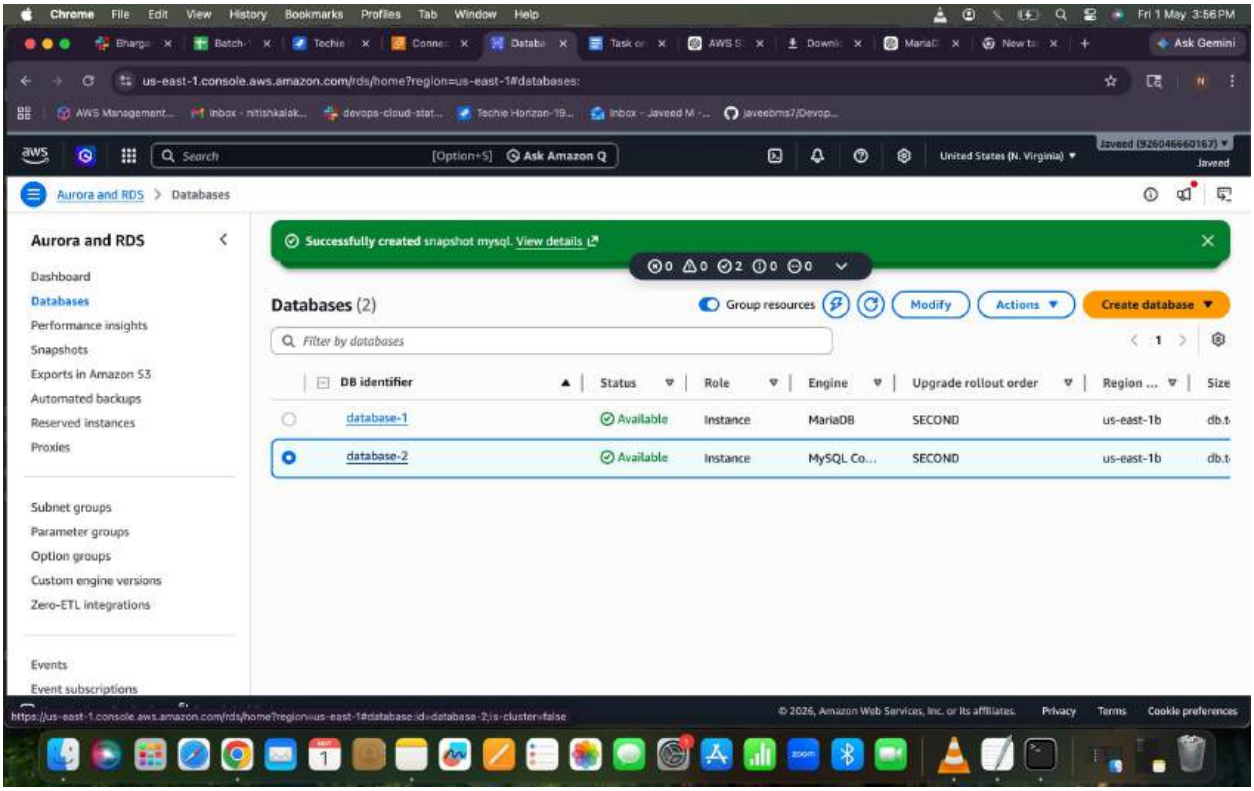
Creating a snapshot



Snapshot Created



Deleting the Database


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 1 May 3:56 PM
us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#databases:
Search [Option+S] Ask Amazon Q
United States (N. Virginia)
Aurora and RDS > Databases
Aurora and RDS
Dashboard
Databases
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Event subscriptions
Successfully created snapshot mysql. View details
Databases (2)
Group resources
Modify
Actions
Create database
Filter by databases
DB identifier
Status
Role
Engine
Upgrade rollout order
Region ...
Size
database-1
Available
Instance
MariaDB
SECOND
us-east-1b
database-2
Available
Instance
MySQL Co...
SECOND
us-east-1b
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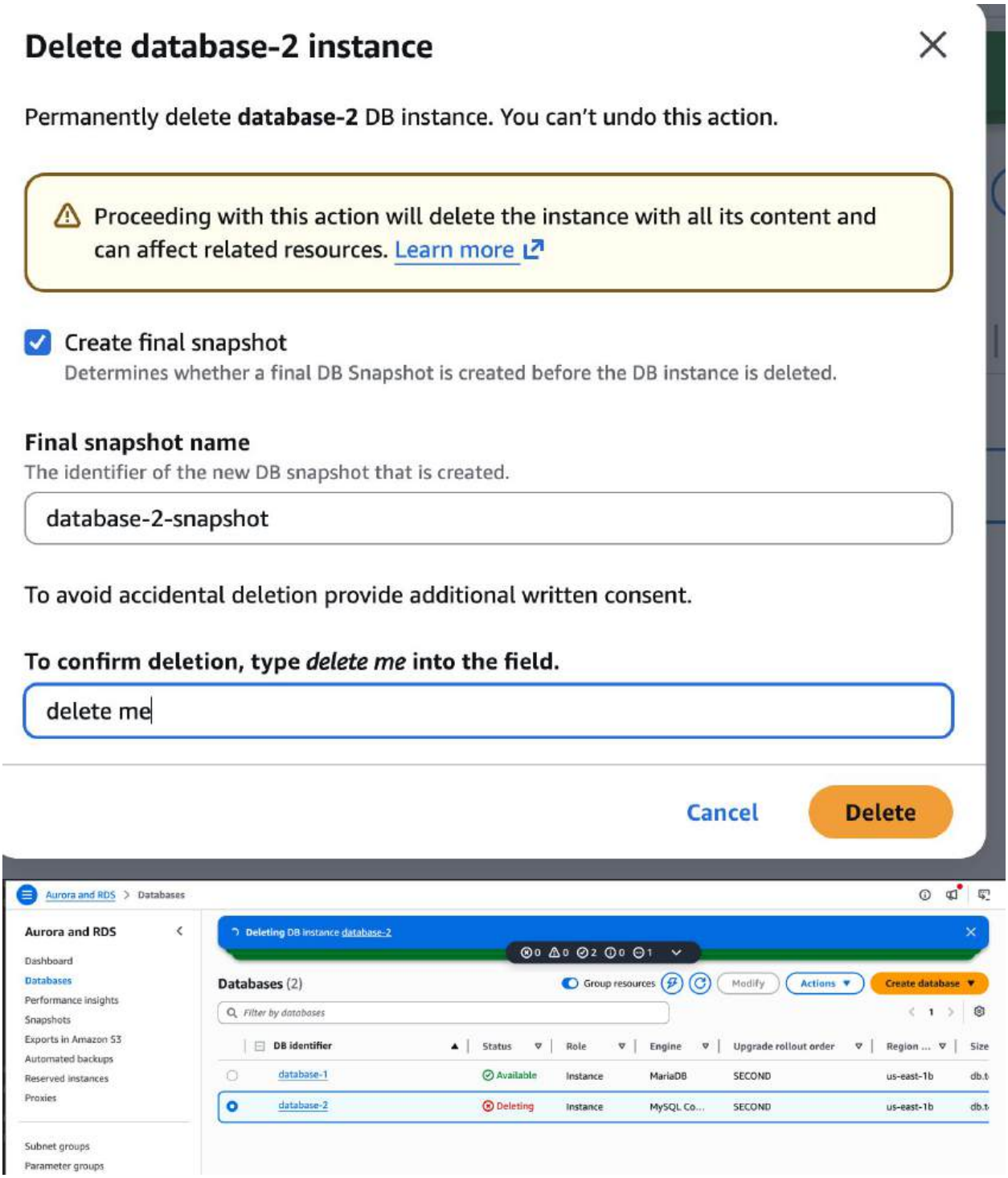
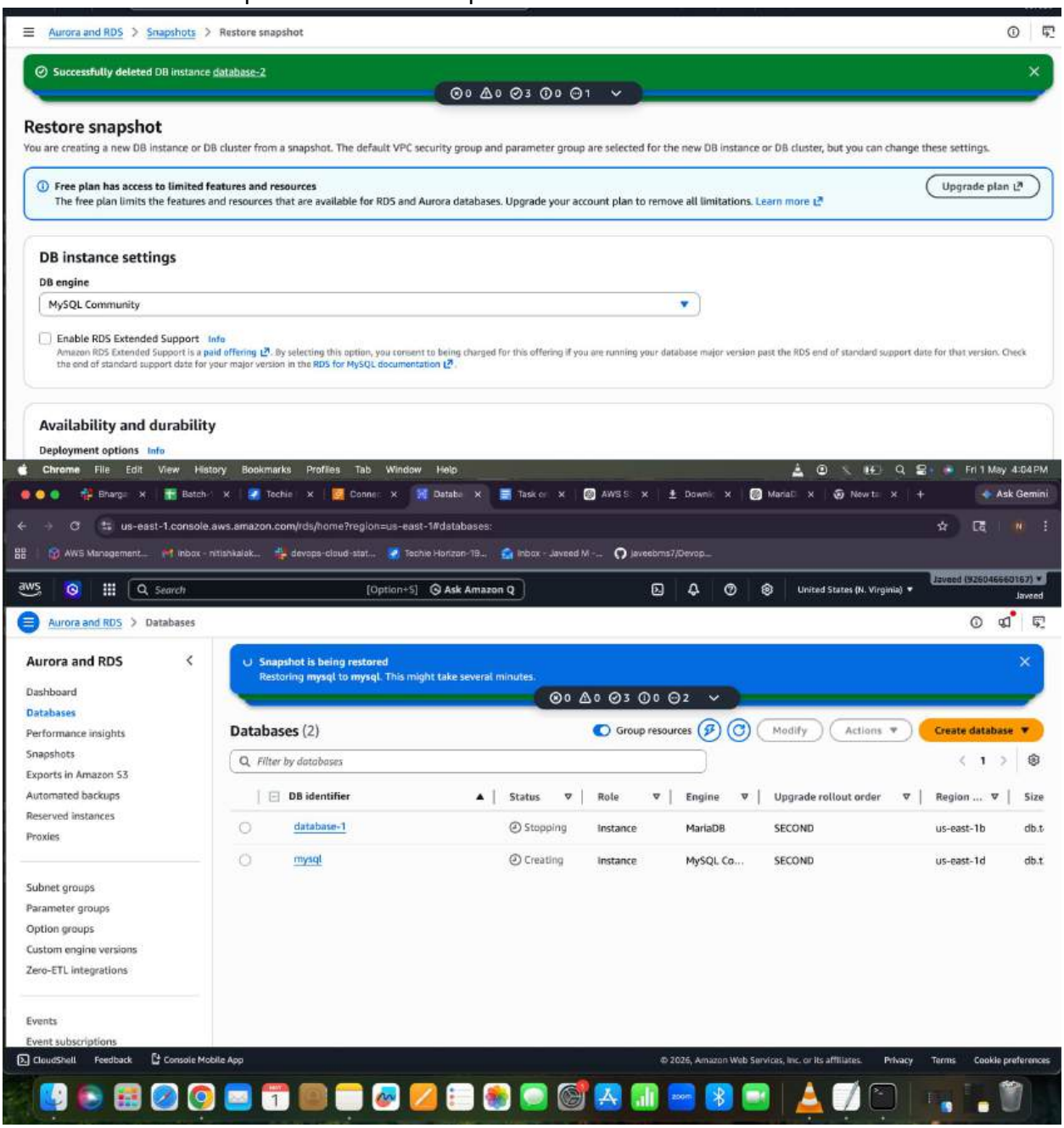
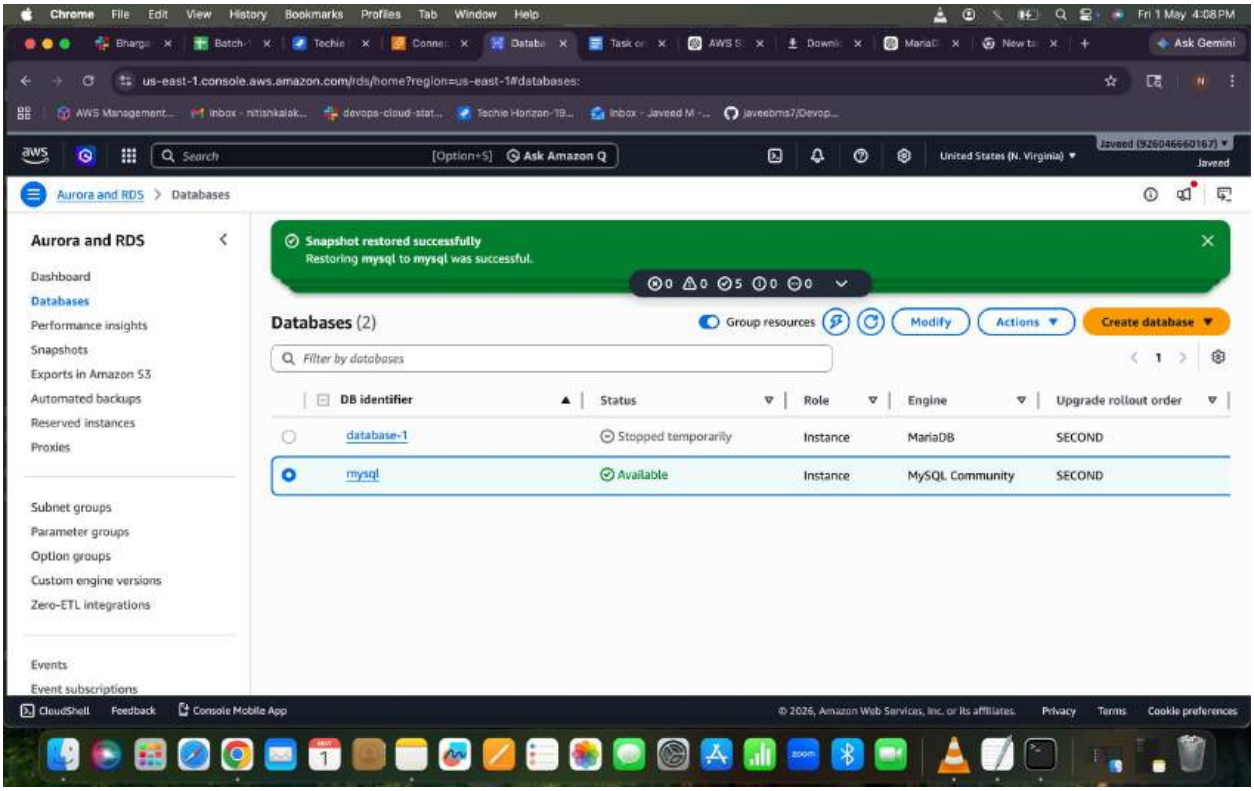


To restore The DB

RDS→Database–Snapshot→Restore Snapshot→create database





10. Create read replica.

To create a read replica we need to enable the backup retention period to at least 1

RDS→Database→Database1→Actions→ Select Create Read Replica→created

Aurora and RDS > Databases > Create read replica

Successfully modified **database-1**.

Create read replica

You are creating a replica DB instance from a source DB instance. This new DB instance will have the source DB instance's DB security groups and DB parameter groups.

Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. Learn more

Upgrade plan

Settings

Replica source
Source DB instance identifier

database-1
Role: Instance

DB instance identifier
This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (for example, mydbinstance).

DB1-Read Replica

Aurora and RDS > Databases

Creating replica DB1-ReadReplica in US East (N. Virginia)
Your database might take a few minutes to launch.

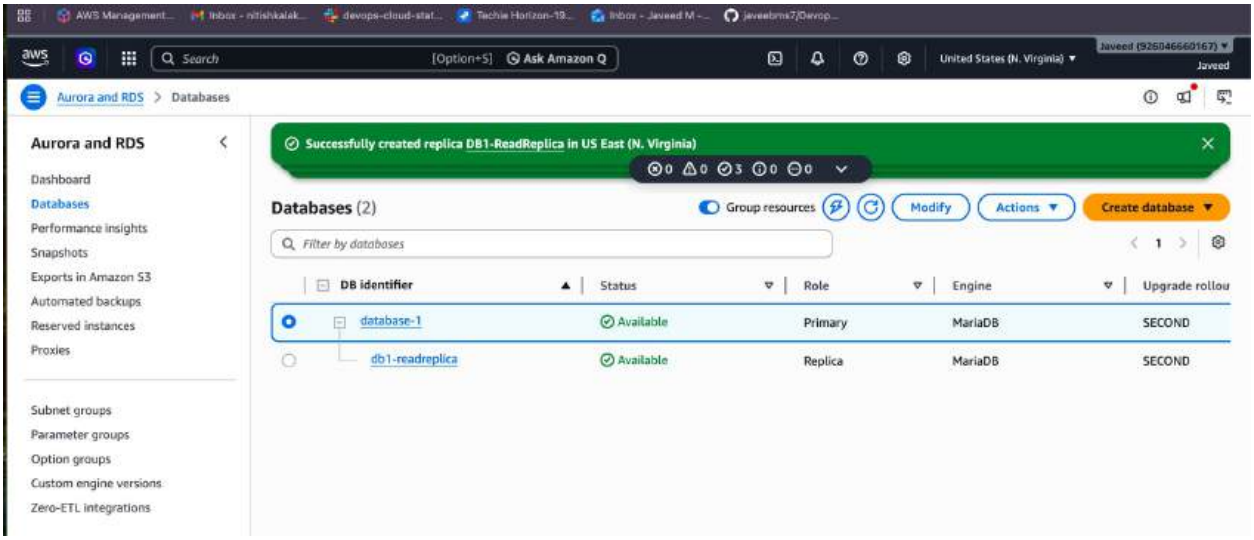
Aurora and RDS
- Dashboard
- Databases
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies
- Subnet groups
- Parameter groups
- Option groups
- Custom engine versions
- Zero-ETL integrations
- Events

Databases (2)

Group resources | Modify | Actions | Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollou
database-1	Modifying	Primary	MariaDB	SECOND
db1-readreplica	Creating	Replica	MariaDB	SECOND



Read Replica Created

Checking on the ec2 instance level we can see the table in my main database has been replicated here

```
[ec2-user@ip-172-31-0-45 ~]$ mysql -h db1-readreplica.cc5o4w40s2bs.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 11.8.6-MariaDB managed by https://aws.amazon.com/rds/

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| ec2db              |
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.01 sec)

mysql> USE ec2db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
```

I tried to create a table in a replica database instance but as it has only read option its throwing up an error.

```
Database changed
mysql> \c
mysql> CREATE TABLE test ( id INT );
ERROR 1290 (HY000): The MariaDB server is running with the --read-only option so it cannot execute this statement
mysql>
```